

TM Forum Component

Service Qualification Management

TMFC009

Maturity Level: General Availability (GA)	Team Approved Date: 18-Aug-2023
Release Status: Preview	Approval Status: Team Approved
Version 1.2.0	IPR Mode: RAND

Notice

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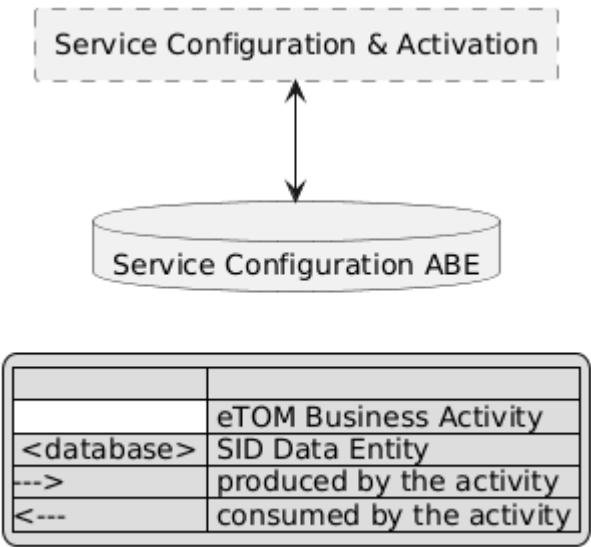
TMFC009 – Service Qualification Management

1. Overview

Component Name	ID	Description	ODA Function Block
Service Qualification Management	TMFC009	The Service Qualification Management component is responsible for checking and validating the availability of a service according to specified and configured business rules. It must identify at least one technical solution (RFSspec) available to deliver the service (CFSspec) and check the availability of all the resources types involved in this technical solution. No resources are allocated during service qualification. The Service Qualification Management component has functionality that includes checking service feasibility status and publishing or reporting service qualification results, but also calculated service delivery due date and identified need of an appointment at the customer site. Service Qualification Management can also be responsible for the cost calculation of the technical solution identified, when it cannot be determined at catalog design time (complex B2B services). This information will be used as an input to price calculation.	Production

2. eTOM Processes, SID Data Entities and Functional Framework Functions

2.1. eTOM L2 - SID ABEs links



2.2. eTOM business activities

eTOM business activities this ODA Component is responsible for:

Identifier	Level	Business Activity Name	Description
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1.4.5	L2	Service Configuration & Activation	Service Configuration & Activation processes encompass allocation, implementation, configuration, activation and testing of specific services to meet customer requirements, or in response to requests from other processes to alleviate specific service capacity shortfalls, availability concerns or failure conditions. Where included in the service provider offering, these processes extend to cover customer premises equipment. / Responsibilities of the Service Configuration & Activation processes include, but are not limited to: • Verifying whether specific service designs sought by customers are feasible as part of pre-order feasibility checks; • Allocating the appropriate specific service parameters to support service orders or requests from other processes; • Reserving specific service parameters (if required by the business rules) for a given period of time until the initiating customer order is confirmed, or until the reservation period expires (if applicable); • Implementing, configuring and activating specific services, as appropriate; • Testing the specific services to ensure the service is working correctly; • Recovery of specific services; • Updating of the Service Inventory Database to reflect that the specific service has been allocated, modified or recovered; • Assigning and tracking service provisioning activities; • Managing service provisioning jeopardy conditions • Reporting progress on service orders to other processes.
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2.3. SID ABEs

SID ABEs this ODA Component is responsible for:

SID ABE L1	SID ABE L1 Definition	SID ABE L2 (or set of BEs)	SID ABE L2 Definition

Service Configuration ABE	The definition of how a Service operates or functions in terms of CharacteristicSpecification(s) and related ResourceSpec(s) and ServiceSpec(s) as well as a representation of how a Product operates or functions in terms of characteristics and related Resource(s) and Service(s). For more detail refer to Configuration & Profiling ABE in Common Domain		(no description available)
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2.4. Functional Framework Functions

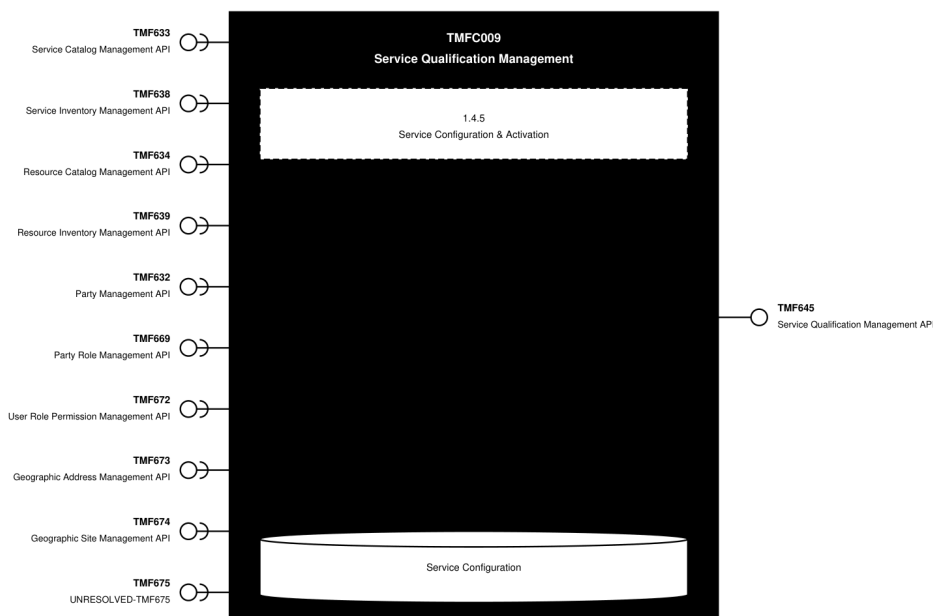
Function ID	Function Name	Aggregate Function L1	Aggregate Function L2	Function Description
319	Service Feasibility Checking	Service Order Management	Service Availability	Service Feasibility Checking provides checking based on the customer service location; service feasibility checks are done to assure the offering can actually be provided to the customer. This implies that the customer location is clearly established. Service feasibility checks are conducted via contract with the Service Order Management function.
586	Service Availability Validation	Service Order Management	Service Availability	Service Availability Validation function validates that the service or services specified on the service order are available at the specified customer/service location and feasible from a network point of view.

571	Service Delivery Due Date Calculation	Service Order Management	Service Order Initialization	Service Delivery Due Date Calculation functions calculates the service delivery due date using network capacity, access provider selection and work center intelligence (including workload and capacity).
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3. TM Forum Open APIs & Events

The following part covers the APIs and Events; this part is split in 3: - List of Exposed APIs - This is the list of APIs available from this component. - List of Dependent APIs - In order to satisfy the provided API, the component could require the usage of this set of required APIs. - List of Events (generated & consumed) - The events which the component may generate are listed in this section along with a list of the events which it may consume. Since there is a possibility of multiple sources and receivers for each defined event.

3.1. API Context Diagram

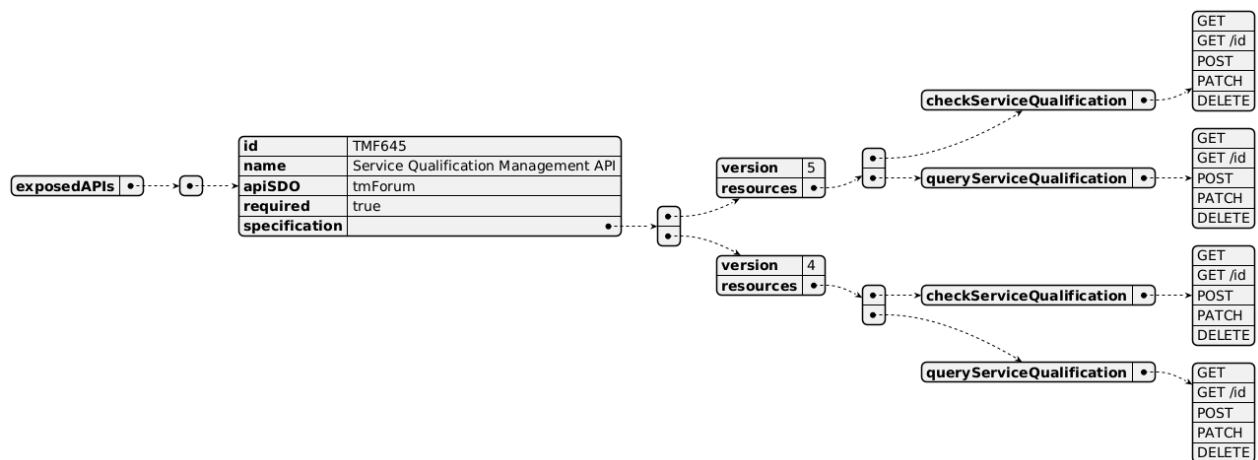


(SVG source: [TMFC009 API Context.svg](#) — hand-drawn rather than PlantUML for this one diagram, since PlantUML's automatic layout couldn't give each API its own straight, individually-anchored connector at a chosen height. Generated by the

component-specification-documentation skill's
scripts/render_api_context_svg.py.)

3.2. Exposed APIs

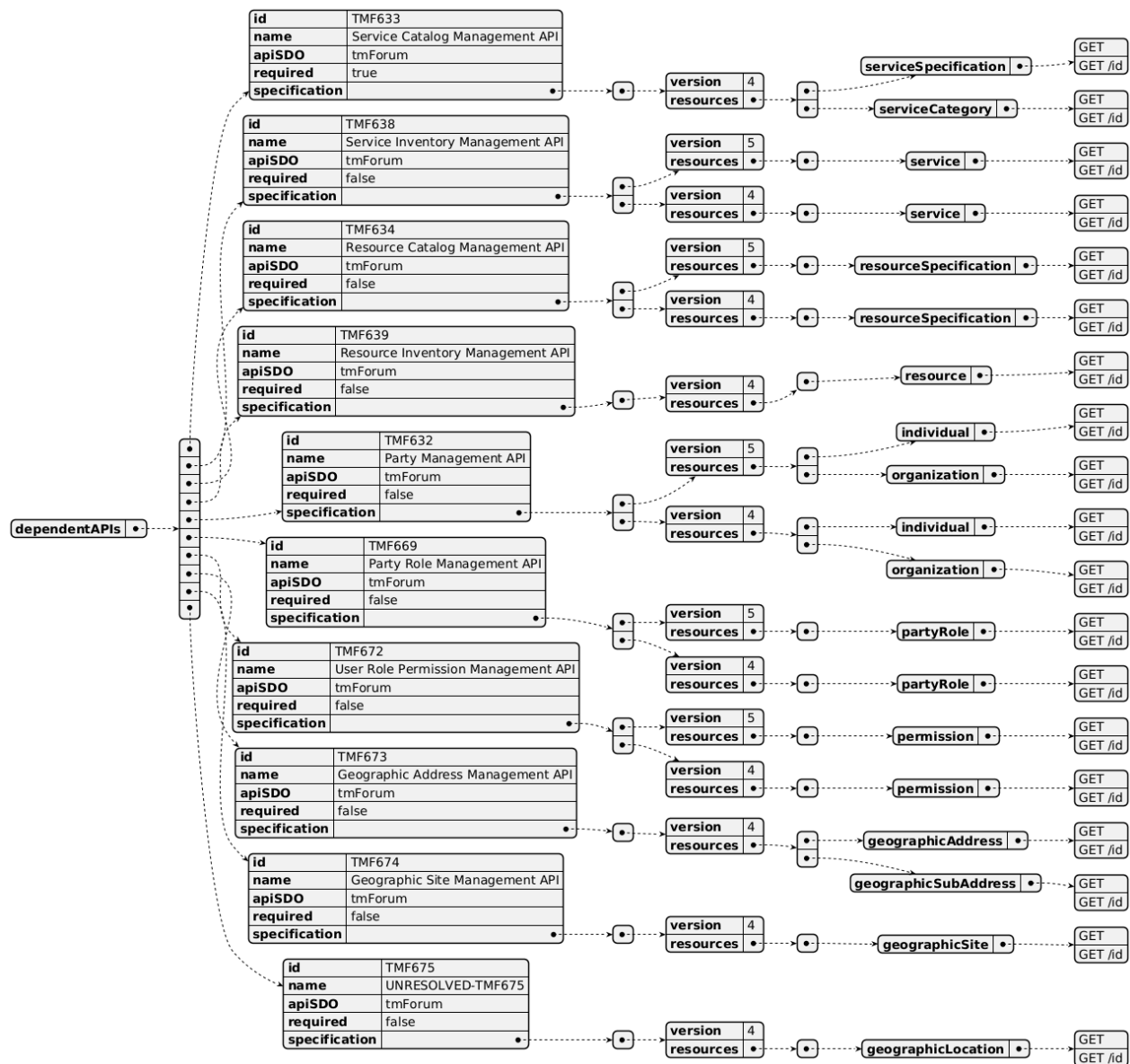
API ID	API Name	Mandatory / Optional	API Version	Resource	Operations
TMF645	Service Qualification Management API	Mandatory	5	checkServiceQualification	GET, GET /id, POST, PATCH, DELETE
TMF645	Service Qualification Management API	Mandatory	5	queryServiceQualification	GET, GET /id, POST, PATCH, DELETE
TMF645	Service Qualification Management API	Mandatory	4	checkServiceQualification	GET, GET /id, POST, PATCH, DELETE
TMF645	Service Qualification Management API	Mandatory	4	queryServiceQualification	GET, GET /id, POST, PATCH, DELETE



3.3. Dependent APIs

API ID	API Name	API Version	Mandatory / Optional	Resource	Operations
TMF633	Service Catalog Management API	4	Mandatory	serviceSpecification	GET, GET /id
TMF633	Service Catalog Management API	4	Mandatory	serviceCategory	GET, GET /id

TMF638	Service Inventory Management API	5	Optional	service	GET, GET /id
TMF638	Service Inventory Management API	4	Optional	service	GET, GET /id
TMF634	Resource Catalog Management API	5	Optional	resourceSpecification	GET, GET /id
TMF634	Resource Catalog Management API	4	Optional	resourceSpecification	GET, GET /id
TMF639	Resource Inventory Management API	4	Optional	resource	GET, GET /id
TMF632	Party Management API	5	Optional	individual	GET, GET /id
TMF632	Party Management API	5	Optional	organization	GET, GET /id
TMF632	Party Management API	4	Optional	individual	GET, GET /id
TMF632	Party Management API	4	Optional	organization	GET, GET /id
TMF669	Party Role Management API	5	Optional	partyRole	GET, GET /id
TMF669	Party Role Management API	4	Optional	partyRole	GET, GET /id
TMF672	User Role Permission Management API	5	Optional	permission	GET, GET /id
TMF672	User Role Permission Management API	4	Optional	permission	GET, GET /id
TMF673	Geographic Address Management API	4	Optional	geographicAddress	GET, GET /id
TMF673	Geographic Address Management API	4	Optional	geographicSubAddress	GET, GET /id
TMF674	Geographic Site Management API	4	Optional	geographicSite	GET, GET /id
TMF675	UNRESOLVED-TMF675	4	Optional	geographicLocation	GET, GET /id



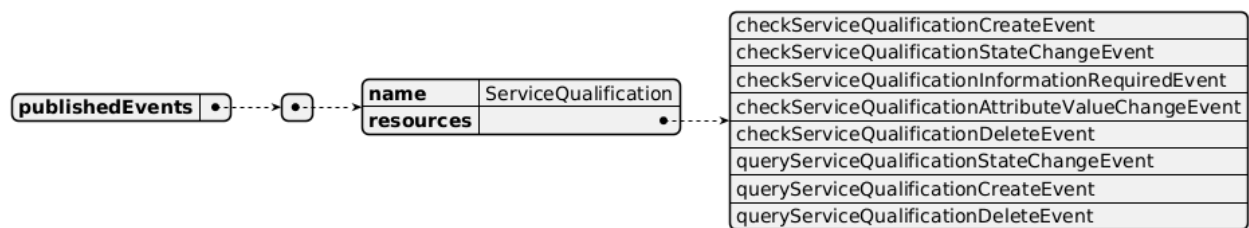
3.4. Events

The diagram illustrates the Events which the component may publish and the Events that the component may subscribe to and then may receive. Both lists are derived from the APIs listed in the preceding sections.

Published Events

API ID	API Name	Event Resources
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TMF645	Service Qualification Management API	checkServiceQualificationCreateEvent checkServiceQualificationStateChangeEvent checkServiceQualificationInformationRequiredEvent checkServiceQualificationAttributeValueChangeEvent checkServiceQualificationDeleteEvent queryServiceQualificationStateChangeEvent queryServiceQualificationCreateEvent queryServiceQualificationDeleteEvent
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Subscribed Events

(none listed in the component YAML)



4. Machine Readable Component Specification

Refer to the ODA Component Directory on the TM Forum website for the machine-readable component specification files for this component.

5. References

5.1. TMF Standards related versions

Standard	Version(s)
eTOM	v23.0.1
SID	v23.5.0
Functional Framework	v23.0

5.2. Jira References

eTOM - Identify a new Service Availability Check/Assessment activity at Service level, as part of L2 Service Configuration & Activation, able to provide all the expected results (feasibility status, service delivery due date, need of an appointment at customer site, and cost of the technical solution when needed) - Restore the Determine Resource Availability activity at Resource level. Initial definition: "This process investigates the ability to be able to satisfy specific service orders as a part of a feasibility check. Where the Allocate & Install Resource processes are requested by a pre-feasibility resource order, or by the Design Resources processes, these processes determine whether the requested resources are available."

SID - Identify a new Service Qualification BE (if we want to register the result of the Service Availability Check/Assessment)

Functional Framework - 592 - Service Parameters Reservation should not be classified in Service Availability sub-domain level 2, but rather in Service Configuration & Activation / Service Configuration - 571 - Service Delivery Due Date calculation should be classified in Service Availability sub-domain level 2 and not in Service Order Initialization - Add a function to calculate the cost of a technical solution, for complex cases when it cannot be done at catalog design time

API - TMF645 (AP-6200 - TMF645 - Add additional information in the response (DONE)) - Service Qualification API: be able to receive as an answer not only the qualification result but also the calculated due date, the need of an appointment, and the cost of the solution identified

5.3. Further resources

As listed in IG1214 - Mapping of ODA Components and ODA Use Cases, this component is involved in the following use cases described in IG1228 - How to Use ODA - Using Open APIs to Realize Use Cases: UC002, UC003, UC007, UC008 and UC010.

6. Administrative Appendix

6.1. Document History

6.1.1. Version History

Version Number	Date	Modified by	Description of changes
1.0.0	05-Aug-2022	Goutham Babu	Final edits prior to publication

1.0.1	25-Jul-2023	Ian Turkington	No content changed, simply a layout change to match template 3. Separated the YAML files to a managed repository.
1.1.0	13-May-2024	Gaetano Biancardi	Component Template Update; ODF 23.5 update; TMF688 removed from exposed and dependent API
1.1.0	12-Jul-2024	Amaia White	Final updates before publication
1.1.1	20-Jan-2026	Rosie Wilson	Updates to description per agreed YAML updated on 03-Jun-2025; final administrative updates prior to publication
1.2.0	29-May-2026	Hugo Vaughan	Gen 5 API Updates
1.2.0	25-Jun-2026	Sannah Sibaya	Final edits prior to publication

6.1.2. Release History

Release Status	Date Modified	Modified by	Description of changes
Pre-production	05-Aug-2022	Goutham Babu	Initial release of document
Pre-production	05-Sep-2022	Adrienne Walcott	Updated to reflect TM Forum Member Evaluated status
Pre-production	14-Aug-2023	Amaia White	New release 1.0.1
Production	06-Oct-2023	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	12-Jul-2024	Amaia White	New release 1.1.0
Production	30-Aug-2024	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	20-Jan-2026	Rosie Wilson	New release 1.1.1
Production	05-Sep-2025	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	25-Jun-2026	Sannah Sibaya	New release 1.2.0

6.2. Acknowledgements

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